

PART 4 — LEVEL 4 (H.C. Verma / JEE ADVANCED) ~120 QUESTIONS

◆ SECTION 1: Electric Charges & Coulomb's Law (Q1–20)

1. Two point charges $+Q$ each are fixed 1 m apart in vacuum. A third charge $-q$ is placed on the line joining them such that net force on it is zero. Find its position.
2. Two point charges $+Q$ and $-2Q$ are separated by distance d . Find the point on line joining them where electric field is zero.
3. A charge $+Q$ is at origin, $-2Q$ at $x = a$. Find electric potential at midpoint.
4. Three charges, q , $2q$, $-q$, placed at corners of right triangle. Find net force on q .
5. Two small identical spheres of mass m suspended from same point by strings of length L . Each carries charge q . Strings form angle θ . Find q .
6. Two charges $+q$ and $-q$ separated by distance $2a$. A third charge $+Q$ placed at midpoint. Find force on it.
7. Four charges $+q$ at vertices of square. Calculate net force on one charge.
8. A charge q is placed at center of equilateral triangle of side a . Electric field at corner:
9. Two charges q_1 , q_2 separated by distance r . Find work done in bringing q_1 from infinity to distance r from q_2 .
10. Two identical charges released from rest at distance d . Find velocity when infinitely far apart.
11. Charges $+Q$ and $-Q$ separated by distance d . Find the work required to swap their positions.
12. Two point charges $q_1 = 1 \mu\text{C}$, $q_2 = 2 \mu\text{C}$ placed 1 m apart in vacuum. Find force. If medium of dielectric constant $K = 4$ inserted, find new force.
13. Charge q is moved along perpendicular bisector of two equal charges. Find force on it as function of distance.
14. Two charges $+q$ each at ends of rod of length L . Find work done in bringing them to this configuration from infinity.
15. A small charge q is placed along line joining two fixed charges Q_1 and Q_2 . Determine condition for stable equilibrium.
16. Two charges $+q$ each, free to move on smooth plane. Initially at distance d . Find velocity when distance doubles due to mutual repulsion.
17. Charge $+Q$ split into two parts q_1 , q_2 at distance r . Total potential energy minimized when $q_1 = q_2$. Show mathematically.
18. Two charges q and $4q$ separated by 1 m. Net force on q if third charge $2q$ placed at midpoint.
19. Three equal charges on line with equal spacing d . Find net force on middle charge.
20. Two charges $+q$ and $-q$ separated by $2a$. A charge Q moves from infinity to midpoint. Find change in potential energy.

◆ SECTION 2: Electric Field & Potential (Q21–40)

21. Electric field at point on axial line of dipole of moment p , distance $r \gg d$: derive formula.
22. Electric field at equatorial line of dipole. Compare with axial field.

23. Potential at point on axis of dipole: derive expression.
24. Potential at point on equatorial plane of dipole: derive.
25. Uniformly charged rod of length L , linear charge density λ . Find electric field at point on axis at distance x from one end.
26. Uniformly charged ring, radius R , total charge Q . Find electric field at point on axis.
27. Uniformly charged ring. Find potential at axial point.
28. Electric field at center of square due to four charges at corners.
29. Electric potential at center of square due to four charges.
30. Electric field due to uniformly charged disk at axial point.
31. Electric potential due to uniformly charged disk at axial point.
32. Electric field at distance r inside uniformly charged solid sphere.
33. Electric field at distance r outside uniformly charged solid sphere.
34. Potential at surface of uniformly charged sphere.
35. Potential inside uniformly charged sphere.
36. Work done in moving a charge along equipotential surface.
37. Equipotential surfaces of a dipole: sketch and explain.
38. Two dipoles along same axis. Find net electric field at midpoint.
39. Charge moves along field line. Find relation between work and potential difference.
40. Two charges $+q$ and $-q$ separated by 4 cm. Find electric field at midpoint and potential.

◆ SECTION 3: Electric Flux & Gauss's Law (Q41–60)

41. A uniformly charged infinite plane sheet. Use Gauss's law to derive electric field.
42. Electric flux through cube enclosing charge Q .
43. Flux through one face of cube.
44. Hollow spherical shell of radius R , charge Q . Find electric field inside and outside.
45. Non-conducting solid sphere, uniform charge density ρ . Find E at distance $r < R$.
46. Electric field of infinite line charge using Gauss's law.
47. Coaxial cylindrical conductor, inner radius a , outer radius b . Find E between conductors.
48. Flux through cylindrical surface enclosing line charge λ .
49. Hollow conductor with cavity. Point charge inside cavity. Find field inside cavity and outside conductor.
50. Electric field just outside surface of charged conductor.
51. Two coaxial spherical shells, inner Q , outer uncharged. Find field in different regions.
52. Field inside conductor in electrostatic equilibrium.
53. Potential inside conductor.
54. Flux through open surface in uniform field.
55. Non-uniform charge distribution in sphere: $\rho \propto r$. Find E using Gauss's law.
56. Thin spherical shell, surface charge density $\sigma(\theta) = \sigma_0 \cos\theta$. Find approximate field at pole.
57. Field at edge of finite sheet.
58. Flux through cube with one vertex at charge Q .
59. Electric field due to charged disk. Compare with field due to plane sheet.

60. Derive expression for field of spherical shell using integration (not only Gauss's law).

◆ SECTION 4: Capacitance, Energy, & Dielectrics (Q61–90)

61. Parallel plate capacitor, area A , separation d , dielectric K . Derive expression for C .
62. Energy stored in capacitor at constant voltage, dielectric inserted.
63. Energy stored at constant charge.
64. Two capacitors in series, connected to battery. Find charges and voltages.
65. Two capacitors in parallel, connected to battery. Find energy stored.
66. Capacitor charged, then disconnected. Dielectric inserted. Find new voltage, charge, energy.
67. Energy density in dielectric medium.
68. Combination of capacitors in complex network. Find equivalent capacitance.
69. Capacitor initially uncharged. Connected in circuit with battery and resistances. Find final charge and energy stored.
70. Two capacitors, one charged, one uncharged. Connected. Find final voltages, charge, energy lost.
71. Capacitance of spherical capacitor. Derive formula.
72. Coaxial cylindrical capacitor. Derive formula.
73. Capacitor with partially inserted dielectric. Find force on dielectric slab.
74. Parallel plate capacitor with slab at angle. Find torque on slab.
75. Two identical charged capacitors connected together. Energy loss calculation.
76. Energy density in parallel plate capacitor in terms of E and ϵ .
77. Electric field distribution inside capacitor with layered dielectrics.
78. Voltage division in series combination.
79. Charge division in parallel combination.
80. Time constant for RC charging of capacitor.
81. Energy stored in capacitor after charging and discharging cycle.
82. Work done in moving charge in capacitor system.
83. Field inside dielectric slab partially inserted in charged capacitor.
84. Derive expression for torque on dipole between capacitor plates.
85. Capacitance of non-uniform plates.
86. Series-parallel network: compute total energy.
87. Dipole near charged plate. Find force and torque.
88. Capacitor network: find maximum energy stored in one capacitor.
89. Charge redistribution problem with dielectrics and multiple capacitors.
90. Energy stored in combination with battery disconnected.

◆ SECTION 5: Dipoles & Advanced Problems (Q91–120)

91. Dipole of moment p in uniform E . Find torque, potential energy at angle θ .
92. Dipole along axial line of another dipole. Find net field at midpoint.

93. Dipole in non-uniform field. Derive expression for force.
94. Two dipoles, perpendicular axes. Find torque and potential energy.
95. Equipotential surfaces of dipole: sketch, derive equation.
96. Field at point on perpendicular bisector of dipole.
97. Charge moves along equatorial line of dipole. Find work done.
98. Dipole near charged plane. Find torque and energy.
99. Electric field at centroid of charged polygon.
100. Potential energy of system of four charges at square corners.
101. Two dipoles placed along same line, separated by distance. Find equilibrium configuration.
102. Electric potential along axis of finite charged rod.
103. Electric field along axis of rod.
104. Work done in assembling multiple charges in geometric configuration.
105. Potential energy of continuous charge distribution.
106. Energy density in field of point charge.
107. Force on dipole in non-uniform field, derive using gradient.
108. Capacitance of composite dielectric system.
109. Charge distribution on isolated conductor.
110. Electric field near surface with surface charge density $\sigma(x,y)$.
111. Parallel plate capacitor with non-uniform separation. Find C.
112. Two capacitors connected: find voltage and charge redistribution.
113. Energy considerations in moving dielectric slab.
114. Torque on dipole at arbitrary orientation in non-uniform field.
115. Multiple charges forming dipole array: compute net potential energy.
116. Capacitance of combination of spherical and cylindrical capacitors.
117. Energy stored in complex RC network.
118. Electric field along axis of charged ring.
119. Electric field at arbitrary point due to system of charges.
120. Derive general expression for energy stored in N-point charges.